Ethical considerations for engineers working in cybernetic implants

Dr Kate Fox
School of Physics
University of Melbourne
Parkville, Victoria, Australia
kfox@unimelb.edu.au

***Abstract*— Research, engineers and ethics are not commonly synonymous. Cybernetic research and particularly neural implant systems exist on the spectra of ethics, especially considering the potential outcome of the research. Neural implants have shown considerable advantages for the repair and replacement of damaged or disabled biological systems and accordingly are rapidly being introduced into mainstream medicine. Concerns however exist over prerequisite preclinical testing and potential misuse of the technology.**

Keywords - engineering; neural implants; ethics; medicine

I. Introduction

Ethics, in principle, is about doing what is right and as an engineer or a scientist, we carry an innate knowledge as to what we believe is defined as right [1]. That is, as an engineer we are taught to use our knowledge to better the universe by building and repairing biological, industrial and mechanical systems. As an individual, we all have our own moral compass whilst as a professional we are guided with a code of conduct provided by the professional societies as to what is considered to be appropriate "professional" behavior. However, simply having codes of ethics does not guarantee ethical behaviour.

Cybernetic research in particular attracts debate due partly to a lack of understanding of the research and due to the influence of outside sources such as the media. Basic biology in the form of reproductive cloning and genetic engineering, for example, has been linked to the formation of a post human society if left unrestricted [2]. Whilst this is somewhat farfetched, the public ethical acceptance of cloning or genetics has been slow. Cybernetics has been met with less ethical objections primarily due to what can only be defined as a "coolness" factor. Cybernetic organisms or cyborgs have been portrayed for decades in science fiction and most people have some understanding of cyborgs through movies like *The Terminator* and TV shows such as *Star Trek*. Accordingly, the concept of combining computing and electronics into medicine and directly interfacing with the human biological system does not seem too foreboding.

II. The baseline for ethical concern

Novel scientific discoveries stemming from cybernetic and neural research bring many basic questions concerning the role of human dignity in medical research and the effect of this research on the long term outlook in society. However, with rapid improvements in research comes increased pressure on the researcher to abide by appropriate ethical guidelines. In order for progression of research towards human clinical trials, most research undergoes a phase of preclinical animal testing. Animal testing has long been an ethical minefield. Although, it is well understood by most scientific researchers that animal trials are one necessary phase for regulatory approval for any implantable medical device, those outside this sphere can find it needless [3, 4].

Human clinical trials as well as the preclinical trials are guided by strict ethical guidelines put in place by both Government agencies and the institution that is hosting the research (committees are preferably comprised of multidisciplinary representatives). Additional ethical consideration is then placed on appropriate selection of candidates and the doctrine of informed consent. When we look into an area such as cybernetics, these two criteria are often challenging to assess. Due to the high risk nature of the body-machine interface, particularly when extended to the brain, the cost of the procedure ensures that careful selection protocols are adhered to ensure that each subject well understands the risk of implantation and the long term and potentially ongoing impact on their lives. Ethically, the researcher must ensure that they manage the subject's expectations. As an example, a clinical trial for deep brain stimulation studies for subjects with motor disorder even with the outcome of significantly improved motor symptoms, remained disappointed by the outcome due to a failure to "meet a preconceived perfect outcome"[5].

III. Cybernetics for the brain-machine interface

Brain-machine interfacing is rapidly presenting new outcomes for the treatment of neural disorders. Recent progress particularly in the capacity to record neural signals open new

978-1-4673-6469-0/13/$31.00 © 2013

therapeutic options for neural prostheses. There are generally three ways to elicit a brain-computer interface, surface electrodes (e.g. EEG), invasive electrodes (e.g. directly contacting the brain) and MRI scanning where commands can be linked to specific areas in the brain. Microelectrode arrays are one means to stimulate cells and record extracellular electrical activity from electrically active neural networks. The development of neural prostheses relies on the aforementioned cycle of preclinical and clinical trials. Basic interfacing issues between tissue and substrate are ironed out without human feedback. Instead non-human models are used for a large proportion of the device assessment and development phases. For example, the extracted brain of a lamprey has been used to control the movement of a robot via light stimulation,[6] insects have been modified with computer chips to redirect their antenna signals to convert the insect into a remote controlled cybernetic insect [7-9] and brain controlled prosthetic arms trialed by monkeys [10, 11].

As an engineer such progress is exciting and it is very simple for enthusiasm to want to speed up the clinical trial. Ethically, it is imperative that the research is progressed in a careful and systematic manner. The primary aim of human-brain interfacing technology is to bring about a level of external control. To achieve this using an animal model can be beneficial to understand the pathways required for simulation, however, the animal model is often unsuitable for any research requiring a level of oral feedback.

The use of neural implants in human subjects remains an infant field. Despite significant bench top and animal trial, it was not until the last 20 years that extended chronic human trials have commenced. Brain-machine interfacing implants have focused for the most part of restoration of function following paralysis. Animal models such as monkeys have been extensively used before human trials commenced on patients with tetraplegia who were able to control a robotic arm after implantation of a 96 electrode array (Fig. 1) [11-13].

A study by Warwick and Ruiz [14] investigated the growing field of neural augmentation systems. Warwick himself a pioneer in cybernetics after a 2002 implantation of a one hundred electrode array into the median nerve fibers of his left arm. Here, however, Warwick and Ruiz discussed the ability of the technological advances to give extra capacity to subjects in the form of a mental upgrade but weighted any ethical discussion and concern within the scope of the extended commercial opportunities [14]. This is a logical conclusion to an Engineer who is already balancing scientific advance with the requisite commercial opportunities. The commercial opportunities to use the ability of computers to replace the less optimally functioning components of the brain for considerable advantage are significant. Ethically however, the brain-machine interface requires significantly more advanced research before highly complex interfacing systems will be allowed into mainstream medicine for anything other than surgical repair. This is due to the aforementioned ethical dilemma of subject expectation[5]. For example, the monkey experiment using a robotic arm by Nicolelis et al.[10] created significant interest within the medical community searching for a solution on viable treatment for paralysis, however, what was not widely reported by the media was that after an 18 month implantation there was a 40% drop in the number of viable electrodes[10, 15].

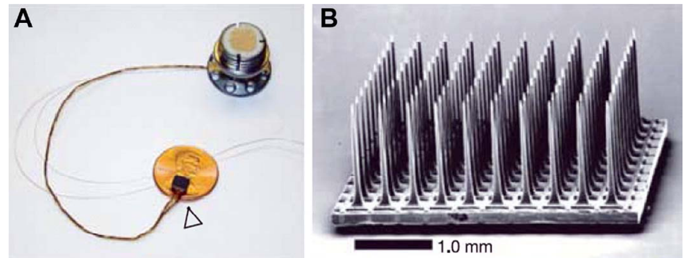

Figure 1: The microelectrode array implanted into the cortex of a tetraplegia patient. Reproduced with permission from [13]

IV. CYBERNETICS FOR THE REPLACEMENT OF ORGANS

Several technological developments in neural engineering have enabled humans to enhance and/or repair their body. To this end, the two organs which have received the most interest are the bionic ear and the bionic eye. Accordingly, neural prostheses have provided the interface and ability to interact directly with nerve cells unable due to loss of function to perform their specified role. Ethically, however, the replacement of body parts has been met with bias. Within the general population, the use of neural implants to replace failing organs appears to be well received provided that it is a like-for-like replacement. Where an implant provides the subject with additional capacity or an "unnecessary" element, approval seems to wane.

The bionic ear, most famously represented as the Cochlear implant, was developed in the later period of the 20th century as a mechanism to restore hearing to the deaf by directly stimulating a multitude of neurons in the cochlea. The cochlear implant comprises a microphone, a speech processor capable of converting sound to electrical signal, a transmission coil, a power supply and a multielectrode array to specifically stimulate the auditory nerve. At the time of genesis the neural implant was considered to be revolutionary as a method to restore a vital sense in subjects who otherwise would not have any other opportunity to regain their hearing. In the last 20 years significant research has been undertaken across the globe in the race for a Bionic Eye predominantly using the lessons learned from the bionic ear. The advancement in Bionic Eye research has come from improvements in the design and fabrication of multielectrode arrays and the communication systems required.

For the most part, retinal prostheses have dominated the bionic eye landscape due to the ease of access and direct contact to the retinal ganglion nerve cells, the optimal cells for direct interfacing. Like the bionic ear, retinal bionic eye prostheses are able to bypass damaged cells and instead electrical stimulating nerve cells.[16-18] Using electrical stimulation, stimulated retinal ganglion cells have been shown to elicit a percept in the form of a phosphene in blind patients [16, 17, 19-21]. Ethically, both the bionic ear and the bionic eye are a minefield. The principles of medical ethics rely on a combination of safety, expectation and efficacy considerations coupled with patient consent and understanding. Considerable debate has occurred in relation to the bionic ear implants, [22, 23] however, bionic eye technology is relatively new and many surgical questions remain. For example, long term efficacy remains undisclosed and even the more established research groups have limited clinical data. Electrode number, device positioning and long term nerve interfacing is not well understood. Even the Argus II retinal implant system from US company Second Sight [17] recently given approval in the United States and Europe suffers from inconsistencies in device positioning and fading of visual perceps [24].

As an engineer it is difficult to see the ethical viewpoints when faced with such exciting technology. Looking to the ethical viewpoints collected for the Cochlear implant and extensively reported it is noticeable that there is considerable debate as to whether deafness is a disability that needs repair [22, 23]. The target market for Cochlear implantation is generally young children. Accordingly, it is difficult to ascertain the ethical standpoint on a parent making the decision on behalf of their child. Stewart-Muirhead [25] in particular suggested that ethical considerations must remain paramount for any new technology stating that “researchers, clinicians and the public must read beyond the excitement of cochlear implant technology and temper enthusiasm with serious ethical reflection”.

V. CYBERNETICS IN THE MILITARY

Modern military medicine has made significant advances into cybernetics largely due to need and an increased availability of money invested into military research. In particular, defense research has provided a scaffold for the capacity to elongate and save lives during and after combat. The United States military has led the way in using technology in their wounded veterans, particularly for the replacement of lost limbs (Fig. 2).

Research such as the i-LIMB, a bionic implant capable of translating the electrical signals generated by the remaining muscles and nerves in the amputated limb into a physical movement by the mechanoelectric arm, has been readily utilised by the military where many wounded soldiers suffering upper-limb amputation wish to return. In order to do so however, the soldiers require use of both arms to show that they are capable of handling their weapons. To this end, in 2011, DARPA, the United States Defense Advanced Research Projects Agency, sought research funding proposals in the aim of returning wounded soldiers to the field [26].

However, these advances in research once again have highlighted our limited knowledge of the research risks and continued to challenge our ethical values. Does the return of these patched-up soldiers give them help or a hindrance? At one side of the debate, the limbs for example can be controlled to move and respond faster than a human, whilst at the other side, researchers have continued to struggle to create a viable link between the owner's motor-control and the limb. Here Grey [27] suggests that “the line between the machine, body and object, has never been so vague as it is in contemporary man-machine weapon systems”

Military medicine has always remained on the borderline of ethics, pushed by direct need and lack of public control. Accordingly, scientific advance is mistakenly perceived as building super-fighting machines. A direct brain interface in the military is one such method towards building the ultimate soldier. Not only does it provide the capacity to return the injured soldier but it presents an opportunity for a soldier to operate outside the confines of biology. For example, certain improvements may enable post-human soldiers to work without the need for rest as well as beyond the limits of human endurance and sensory abilities[28]. It is the scientists and engineers that need to be part of the debate to ensure the progress to all research that investment in military advances will impart.

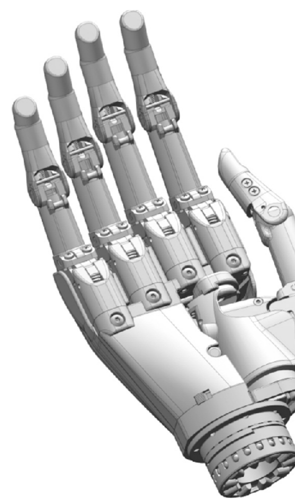

Figure 2: An example of a bionic limb. Reproduced with permission from [29]

VI. TREATMENT VERSUS IMPROVEMENT

As an engineer, we do not see the ethical line between neural interfaces designed to treat or improve. However, this is often where the ethics is divided. As an engineer, we seek to design and implement the best technology possible without much consideration about the outcome. Funding is usually based upon techniques that will improve lifestyles, however, after generation of the research, it is rare that the researcher retains control over how their published work is subsequently developed or used. Cybernetics and neural implants can take a variety of forms from replacement of biological systems to improved functionality. The engineer’s goal is to enhance capabilities and performance and we do not control whether it is used for a body ravaged by disability or just for cosmetic improvement.

Referring to Warren et al.[30], ethics is comprised of a number of intertwined elements (Fig. 3). Safety for example is an obvious decision point; however, with limited long term data the safety of the cybernetic implant is often unknown. Freedom of choice places the decision into the hands of the recipient and highly dependent on the information presented to the subject whilst coercion, deliberate or otherwise is clearly ethically understood by doctors, engineer and subjects. The interesting considerations lie in the elements of character and justice. First, character asks the ethical question or whether the improvement is cheating, for example, how does society expect the subject to respond to the implant. That is, will the implant give the subject an unfair advantage in a society that expects subjects to toil for benefit (no pain no gain)[30] and where great value is placed into the notion of personhood (changing one element will change your value as a human)[31]. This is where treatment becomes more "ethically" viable as one seeks repair to rebuild the human self. This is somewhat misguided.

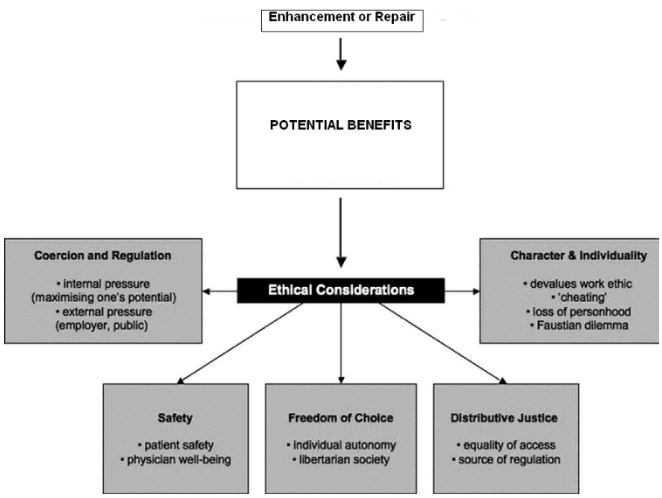

Figure 3: The ethical landscape. Modified from [30]

The last ethical element relies on distribution, that is, is this cybernetic treatment selective and only available to the rich and preferred. As an engineer, we research on the basis of the human good hoping that our technology will be available to all. As a business, however, research requires return on investment.

VII. CONCLUSION

Cybernetics and neural implants are undergoing a period of rapid improvement. Interfacing biology with machine provides society with a vast toolbox of methods to advance, treat and repair the human body. However, will great advance introduces new ethical questions. Engineers working in the field often remain shielded from societal pressure into the appropriate use of their research and more so, are often unable to prevent use of their research outside the initial scope for which it was created. Ethics in principle exists on the moral line of doing what is right and accordingly, I have no concerns about the role the engineer and the scientist play in the research landscape.

ACKNOWLEDGMENT

K. Fox acknowledges the support of a Melbourne Materials Institute seeding grant.

[1] H. Thimbleby, "Robot ethics? Not yet A reflection on Whitby's ''Sometimes it's hard to be a robot''," *Interacting with Computers,* vol. 20 pp. 338-341, 2008.

[2] F. Fukuyama, *Our Posthuman Future. Consequences of the Biotechnology Revolution*. New York: Farrar Straus and Giroux, 2002.

[3] J. A. Smith and K. M. Boyd, *Lives in the Balance: The Ethics of Using Animals in Biomedical Research*. Oxford: Oxford University Press, 1992.

[4] A. N. Rowan., "Formulation of ethical standards for use of animals in medical research," *Toxicology Letters,* vol. 68, pp. 63-71, 1993.

[5] M. Okun and K. Foote, "A mnemonic for Parkinson disease patients considering DBS: a tool to improve perceived outcome of surgery.," *Neurologist,* vol. 10, p. 290., 2004.

[6] K. F. B. Reger, V. Sanguineti, S. Simon Alford, F. Mussa-Ivaldi, "Connecting brains to robots: an artificial body for studying computational properties of neural tissues," *Artif. Life,* vol. 6 pp. 307-324, 2000.

[7] T. Latif and A. Bozkurt, "Line following terrestrial insect biobots " *34th International Conference of the IEEE Engineering in Medicine and Biology Society,* pp. 972 - 975, 2012.

[8] A. Bozkurt, R. F. Gilmour, A. Sinha, D. Stern, and A. Lal, "Insect–Machine Interface Based Neurocybernetics " *IEEE Transactions on Biomedical Engineering,,* vol. 56, pp. 1727 - 1733, 2009.

[9] A. Bozkurt, R. F. Gilmour, and A. Lal, "Balloon-Assisted Flight of Radio-Controlled Insect Biobots " *IEEE Transactions on Biomedical Engineering,,* vol. 56, pp. 2304 - 2307, 2009.

[10] D. D. M. Nicolelis, J. Carmena, R. Crist, G. Lehew, J. Kralik, S. Wise, "Chronic, multisite, multielectrode recordings in macaque monkeys," *Proc. Natl. Acad. USA,* vol. 100, pp. 11041-11046, 2003.

[11] S. Perel., A. B. Schwartz., M. C. Spalding., M. Velliste., and A. S. Whitford., "Cortical control of a prosthetic arm for self-feeding," *Nature,* vol. 453, pp. 1098-1101, 2008.

[12] E. E. Fetz., C. T. Moritz., and S. I. Perlmutter., "Direct control of paralysed muscles by cortical neurons," *Nature,* vol. 456, pp. 639-643, 2008.

[13] L. R. Hochberg, M. D. Serruya, G. M. Friehs, J. A. Mukand, M. Saleh, A. H. Caplan, A. Branner, D. Chen, R. D. Penn, and J. P. Donoghue, "Neuronal ensemble control of prosthetic devices by a human with tetraplegia," *Nature,* vol. 442, pp. 164-171, 2006.

[14] K. Warwick. and V. Ruiz., "On linking human and machine brains," *Neurocomputing,* vol. 71, pp. 2619-2624, 2008.

[15] V. S. Polikov, P. A. Tresco, and W. M. Reichert, "Response of brain tissue to chronically implanted neural electrodes," *Journal of Neuroscience Methods,* vol. 148, pp. 1-18, 2005.

[16] M. S. Humayun, E. de Juan Jr, G. Dagnelie, R. J. Greenberg, R. H. Propst, and D. H. Phillips, "Visual perception elicited by electrical stimulation of retina in blind humans," *Arch Ophthalmol,* vol. 114, pp. 40-6, Jan 1996.

[17] M. S. Humayun, J. D. Dorn, L. da Cruz, G. Dagnelie, J. A. Sahel, P. E. Stanga, A. V. Cideciyan, J. L. Duncan, D. Eliott, E. Filley, A. C. Ho, A. Santos, A. B. Safran, A. Arditi, L. V. Del Priore, and R. J. Greenberg, "Interim results from the international trial of Second Sight's visual prosthesis," *Ophthalmology,* vol. 119, pp. 779-88, Apr 2012.

[18] A. E. Hadjinicolaou, R. T. Leung, D. J. Garrett, K. Ganesan, K. Fox, D. A. X. Nayagam, M. N. Shivdasani, H. Meffin, M. R. Ibbotson, S. Prawer, and B. J. O'Brien, "Electrical stimulation of retinal ganglion

cells with diamond and the development of an all diamond retinal prosthesis," *Biomaterials,* vol. 33, pp. 5812-5820, 2012.

[19] E. Zrenner, H. Benav, A. Bruckmann, U. Greppmaier, A. Kusnyerik, A. Stett, K. Stingl, and R. Wilke, "Electronic Implants Provide Continuous Stable Percepts in Blind Volunteers Only if the Image Receiver is Directly Linked to Eye Movement," *ARVO Meeting Abstracts,* vol. 51, p. 4319, April 11 2010.

[20] E. Zrenner, K. U. Bartz-Schmidt, F. Gekeler, U. Greppmaier, S. Hipp, G. Hoertdoerfer, C. Kernstock, A. Kusnyerik, H. Sachs, and K. Stingl, "Seeing With Subretinal Electronic Implants: Study in Ten Patients With Wireless Implant Alpha-IMS," *ARVO Meeting Abstracts,* vol. 53, p. 6948, March 26 2012.

[21] E. Zrenner, K. U. Bartz-Schmidt, H. Benav, D. Besch, A. Bruckmann, V. P. Gabel, F. Gekeler, U. Greppmaier, A. Harscher, S. Kibbel, J. Koch, A. Kusnyerik, T. Peters, K. Stingl, H. Sachs, A. Stett, P. Szurman, B. Wilhelm, and R. Wilke, "Subretinal electronic chips allow blind patients to read letters and combine them to words," *Proceedings. Biological sciences / The Royal Society,* vol. 278, pp. 1489-97, May 22 2011.

[22] T. P. Gonsoulin, "Cochlear implant/Deaf World dispute: Different bottom elephants," *Otolaryngology - Head and Neck Surgery,* vol. 125, pp. 552-556, 2001.

[23] T. Balkany, A. Hodges, and K. Goodman, "Ethics of cochlear implantation in young children," *Otolaryngology - Head and Neck Surgery,,* vol. 114, pp. 748-755, 1996.

[24] A. Perez Fornos, J. Sommerhalder, L. da Cruz, J. A. Sahel, S. Mohand-Said, F. Hafezi, and M. Pelizzone, "Temporal properties of visual perception on electrical stimulation of the retina," *Invest Ophthalmol Vis Sci,* vol. 53, pp. 2720-31, May 2012.

[25] E. Stewart-Muirhead, "'Fixing' deafness: Ethical issues in cochlear implantation," *http://gpu.srv.ualberta.ca/~ethics/bb6-4fix.htm,* 2000.

[26] Microsystems Technology Office., "Broad Agency Announcement: Reliable Central-Nervous-System Interfaces (RCI)," March 30, 2011

[27] C. H. Gray, *Postmodern war*. London, 1997.

[28] C. Masters, "Cyborg soldiers and militarised masculinities," *International Feminist Journal of Politics,* vol. 7, pp. 112-132, 2005.

[29] R. G. E. Clement, K. E. Bugler, and C. W. Oliver, "Bionic prosthetic hands: A review of present technology and future aspirations," *The Surgeon,* vol. 9, pp. 336-340, 2011.

[30] O. J. Warren, D. R. Leff, T. Athanasiou, C. Kennard, and A. Darzi, "The Neurocognitive Enhancement of Surgeons: An Ethical Perspective," *Journal of Surgical Research,* vol. 152, pp. 167-172, 2009.

[31] A. Chatterjee, "Cosmetic neurology: The controversy over enhancing, movement, mentation, and mood," *Neurology,* vol. 63, p. 968, 2004.